Designing clinical research

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Clinical Research

- Start with a well formulated question
- Choose the study design (that best answers your question)
- Define the population and variables
- Measure these variables
- Analyze the results

Essential documents

- Protocol
- Curriculum Vitae and Licenses
- Data collection Form
- Informed Consent Form
- IRB Approval

The protocol

- Background and rationale
- Design
- Participants: Selection criteria
- Variables: Predictor, Confounding, Outcome
- Statistical issues: Hypotheses, Sample size, Analytic approach

Today, we shall learn how to

- ***Calculate*** the size of the study
- ***Prepare*** a data collection form

Sample Size and Power

The key to healthy urinary tract

- Cranberry has a high content of vit C, fiber, antioxidants
- Cranberry may prevent UTIs
- Cranberry
  - Sauce
  - Juice
  - Capsules

What I want to show

- Consuming cranberry promotes healthy urinary tract
- How many participants do I need to answer the question?

Ingredients for Estimating Sample Size

- Testable hypothesis
- Study design
- Effect size
- Power and alpha

Sample Size Ingredients

- **Testable hypothesis**
- Study design
- Effect size
- Power and alpha

My research question

- Is consuming cranberry correlated with a healthy urinary tract?
- What is wrong?

What’s wrong?

- Vague
- Not testable
- Not measurable
- This is not the correct use of the term “correlation”

“Consuming cranberry”

- Needs to be quantifiable
- I could measure cranberry intake in a sample, then follow-up for an outcome ……

An alternative to “Consuming cranberry”

- A known quantity of cranberry
- Juice or sauce intake would be difficult
- Therefore, a specific, measured, predictor
  - cranberry supplemental capsules with a specific dose

I need a measurable endpoint

- “improves urinary tract health”
  - Not measurable
- A measurable endpoint: UTI

The question is getting better

- Do “people” who get a cranberry supplement have a lower UTI than those who do not get the supplement?

But we must Specify the population

- Women during pregnancy
  - Because women have a relatively higher risk of UTI
  - Because pregnancy increases the risk of UTI:
    - Enough UTI within a feasible period of study

The research hypothesis

- Women during pregnancy who get a cranberry supplement have a lower UTI than those who do not get the supplement.
  - This is the‘alternative’ hypothesis
  - It cannot be tested statistically
- Statistical tests only reject null hypothesis

The Null Hypothesis

- Women during pregnancy who get a cranberry supplement do NOT have a lower UTI than those who do not get the supplement.

Sample Size Ingredients

- Testable hypothesis
- **Study design**
- Effect size
- Power and alpha

Descriptive studies

- Sample size is based on the desired width of the confidence interval
- What is the proportion of women who suffer UTI during pregnancy?
  - Confidence interval for the proportion

Analytical studies

- supplement vs no supplement
  - Null hypothesis: $p1 = p2$ or $P1 – p2 = 0$

Sample Size Ingredients

- Testable hypothesis
- Study design
- **Effect size**
- Power and alpha

Estimating the effect size

- Start with the baseline risk (proportion in the unexposed, those who do not take supplement)
- Usually available from prevalence studies
- The proportion of women who develop symptomatic UTI during pregnancy is 10%

The effect of cranberry on risk of UTI

- What should I assume for the effect of cranberry in preventing UTI?
- Ways to choose an effect size:
    - What is likely, based on other data?
    - Do a pilot study
    - What difference is important to detect?
        - I don't want to miss a 2% decrease
    - What can we afford?
        - 2%: the trial will be too big & expensive
        - 5%: the trial will be smaller and cheaper

Sample Size Ingredients

- Testable hypothesis
- Study design
- Effect size
- **Power and alpha**

α (alfa): false positive (Type I error)

- Dogma: **α** (alpha) = 0.05
- If I need to convince skeptics, I shall reduce the chance of a false positive
  - Very small chance that a positive result is an error
  - **α** (alpha) = 0.01
  - A smaller **α** means larger sample size

Two-sided vs. one-sided **α**

- A two-sided **α** assumes that the result could go either way
  - Cranberry decreases UTI
  - Cranberry increases UTI
- A one-sided hypothesis assumes that the result could, plausibly, go only one way

One-sided **α**

- You may believe that your effect could only go one way!
  - Cranberry is ‘natural.’ It could not increase UTI!
- Be humble
  - The history of research is filled with results that contradicted expectations
  - Remember fluoride and osteoporosis
  - A one-sided test is almost never the best choice

ß (beta): false negative (Type II error)

- Dogma ß (beta) = 0.2
  - The probability of missing the effect (ß)
- Power (1- ß)
  - The **probability** of rejecting the null hypothesis when a specific alternative hypothesis is true.
  - Power = 1- ß = 0.80

I really don't want to miss it

- ß = .10
- Power (1- ß) = 0.90
- Greater power requires a larger sample size

We have all the ingredients

- Testable hypothesis
- Study design: analytical
- Effect size 10% vs. 5%
- Power: 0.8
- alpha: two-sided 0.05

How to get the estimates

- Use R
- Online sample size calculators

Common sample size mistakes

- not performing any calculations
- selecting sample size based on convenience
- making unrealistic assumptions
- failing to account for potential losses during the study

Data collection form

Data Collection Form

- Case Report Form
- Questionnaire
- Interview: Prompts and Responses
- Instrument e.g., SF-36 (36-item Short-Form Health Survey)
- Medical Record Abstraction Form

Data Collection Forms

- is Mandatory to validate and replicate findings
- must reflect the protocol (i.e. Collect the variables of the study)
- must be developed and tested in advance
- Is either a printed or electronic document.
  - Paper data capture → Keyboard Transcription
  - Electronic data Capture

Advantages of electronic Forms

- Avoid the transcription step
- Include validation checks
- Incorporate skip logic

Web-Based Data Collection Forms

- MS forms
- Qualtrics
- REDCap

Principles: Clarity is the keyword

- The cover page should contain
  - Consent check box
  - Eligibility criteria – As a check list
- Clear and concise questions, prompts, and instructions
- Clear guidance about skip patterns
- Use Visual cues
- Provide pre-coded answer options

Principles: Clarity is the keyword

- Avoid free text fields, where possible
- Collect raw data not calculated data
- a field (column) must contain only one type of data
- Specify the unit of measurement
- Indicate the number of decimal places

Raw data, not calculated data

- Weight in kg, Height in cm, not BMI
- DOB and Date of Measurement, not Age
- Race, Sex, Creatinine, not eGFR.

Fields

- a field (column) must contain only one type of data
- Wrong: BP field
- Correct: two fields, one for SBP and another for the DBP

Yes/No fields

- Answers are coded as 1 = Yes and 2 = No.
- If the codes are assigned in this order, the same order should be practiced throughout the CRF
- Be careful. In certain questions “No” may not be distinguished from blank.
- Use option group with radio buttons and allow “Don’t Know”

MECE fields

- Mutually exclusive, collectively exhaustive choices are displayed as an “option group” of “radio buttons” or as a dropdown list.
  - Mutually Exclusive: e.g., age groups 0–10 and 11–20 do not overlap.
  - Collectively Exhaustive: e.g. categories of {<20, 20–40, >40} cover all ages.
- Choices which are not mutually exclusive are displayed as check boxes e.g. co-morbidity: HTN, DM, etc

Data Dictionary

- Field Name
- Data Type (number, text, date)
- Description/Label
- Units (when appropriate)
- Response options
- Meaning of response codes (when appropriate)
- Validation rules

A poorly designed form

- Date of visit: ___________
- Blood pressure: _____ / _____
- Pulse: _____
- Temperature: _____
- Respiration: _____

Well designed form Plus coding

Data of Birth (DD/MM/YYYY)	□□/ □□/ □□ □□
Gender	M □ 1 F □ 2
Height (cm)	□□□.□
Weight (Kg)	□□□.□□
Smoker	Yes □ 1 No □ 2
Sexually active	Yes □ 1 No □ 2 NA □ 9

Test Your Form!

- Fill it out yourself (and time yourself)
- Have your team members fill it out (time them)
- Try it out on a potentially eligible participant (time them)

Ideas

Brainstorming